3 Fast Fourier Transform

The Fast Fourier Transform (FFT) is an algorithm that efficiently computes the Discrete Fourier Transform (DFT). The DFT of a sequence {x(n)} of length N is given by a complex X-valued sequence {X(K)}.

$$X(K) = \sum_{n=0}^{N-1} x(n) W_N^{kn}, \; K = 0, 1, \ldots N\text{-}1$$

Where, $W_N = e^{-j2\pi N}$

In general, the data sequence x(n) is also assumed to be complex valued.

$$x(n) = \frac{1}{N} \sum_{n=0}^{N-1} X(K) W_N^{-Kn}, \; n = 0, 1, \ldots. N\text{-}1$$

Note:

1. For each value of K, direct computation of X(k) involves N complex multiplications (4N real multiplications) and N-1 complex additions (4N – 2 real additions).
   Each complex multiplication requires 4 real multiplications and 2 real additions and each complex addition requires 2 real additions. Therefore, for each values of K, the number of real multiplications = 4N, number of real additions = 2N + 2(N – 1) = 4N –2
2. To compute all the values of DFT requires N^2 complex multiplications and $N^2 – N$ complex additions.

Direct computation of the DFT is basically inefficient primarily because it does not exploit the symmetry and periodicity properties of the phase factor W_N.

Symmetry property: $W_N^{K+\frac{N}{2}} = -W_N^K$

Periodicity property: $W_N^{K+N} = W_N^K$

Direct Computation of the DFT

If x(n) is a complex valued signal

$x(n) = x_R(n) + j\, x_I(n)$

The N-point DFT can be expressed as

$$X(k) = X_R(k) + j\, X_I(k)$$

$$= \sum_{n=0}^{N-1} \left[x_R(n) + j x_I(n)\right] \left[\cos\frac{2\pi kn}{N} - j\sin\frac{2\pi kn}{N}\right]$$

$$X_R(k) + j\, X_I(k) = \sum_{n-0}^{N-1} \left[x_R(n)\cos\frac{2\pi kn}{N} + x_I(n)\sin\frac{2\pi kn}{N}\right]$$

$$- j\left[x_R(n)\sin\frac{2\pi kn}{N} - x_I(n)\cos\frac{2\pi kn}{N}\right]$$

$$X_R(k) = \sum_{n=0}^{N-1} \left[x_R(n)\cos\frac{2\pi kn}{N} + x_I(n)\sin\frac{2\pi kn}{N}\right] \quad \ldots (i)$$

$$X_I(k) = -\sum_{n=0}^{N-1} \left[x_R(n)\sin\frac{2\pi kn}{N} + x_I(n)\cos\frac{2\pi kn}{N}\right] \quad \ldots (ii)$$

The direct computation of equations (i) and (ii) requires

- $2N^2$ evaluations of trigonometric functions.
- $4N^2$ real multiplications.
- 4N (N – 1) real additions.
- A number of indexing and addressing operations.

These indexing and addressing operations are necessary to fetch the data x(n), $0 \le n \le N - 1$ and the phase factors and to store results.

Decimation in Time FFT Algorithms

To increase efficiency, it is necessary to decompose the DFT computation into successive smaller DFT computation.

In this process, we exploit both the symmetry and the periodicity of complete exponential $W_N^{kn} = e^{-j\frac{2\pi kn}{N}}$.

The decimation in time FFT algorithm is based on splitting (decimating) x(n) into smaller sequences and finding X(k) from the DFT's of these decimated sequences.

Let x(n) be a sequence of length $N = 2^v$.

Since N is an even integer, x(n) is now split into two $\frac{N}{2}$ points sequence consisting of the even numbered points in x(n) and odd number points in x(n).

$$X(K) = \sum_{n=0}^{N-1} x(n) W_N^{kn}, \; K = 0, 1, \ldots.. N - 1$$

Separating x(n) into its even and odd number points.

$$X(K) = \sum_{n\,\text{even}} x(n) W_N^{kn} + \sum_{n\,\text{odd}} x(n) W_N^{kn}$$

Put n = 2r for n even

n = 2r + 1 for n odd

$$X(K) = \sum_{r=0}^{\left(\frac{N}{2}-1\right)} x(2r) W_N^{2rk} + \sum_{r=0}^{\frac{N}{2}-1} x(2r+1) W_N^{(2r+1)k}$$

$$= \sum_{r=0}^{\frac{N}{2}-1} x(2r)\left(W_N^2\right)^{rk} + W_N^k \sum_{r=0}^{\frac{N}{2}-1} x(2r+1)\left(W_N^2\right)^{rk}$$

$$W_N^2 = e^{-j2\left(\frac{2\pi}{N}\right)} = e^{-j2\pi/N/2} = W_{N/2}$$

$$X(K) = \sum_{r=0}^{\frac{N}{2}-1} x(2r) W_{N/2}^{rk} + W_N^k \sum_{r=0}^{\frac{N}{2}-1} x(2r+1) W_{N/2}^{rk}$$

$$X(K) = G(K) + W_N^k H(K) \quad \text{... (i)}$$

Equation (i) is recognised as an $\frac{N}{2}$ point DFT, the first sum being $\frac{N}{2}$ point DFT of the even numbered points of the original sequence and the second being the $\frac{N}{2}$ point of the odd numbered points of the original sequence.

$$X\left(K+\frac{N}{2}\right)$$

$$= \sum_{r=0}^{\frac{N}{2}-1} x(2r) W_{\frac{N}{2}}^{r\left(K+\frac{N}{2}\right)} + W_N^{\left(K+\frac{N}{2}\right)} \sum_{r=0}^{\frac{N}{2}-1} x(2r+1) W_{\frac{N}{2}}^{r\left(K+\frac{N}{2}\right)}$$

$$= \sum_{r=0}^{\frac{N}{2}-1} x(2r) W_{\frac{N}{2}}^{rk} W_{\frac{N}{2}}^{r\frac{N}{2}} + W_N^K \cdot W_{\frac{N}{2}}^{r\frac{N}{2}} \sum_{r=0}^{\frac{N}{2}-1} x(2r+1) W_{\frac{N}{2}}^{rK} W_N^{N/2}$$

$$X\left(K+\frac{N}{2}\right) = \sum_{r=0}^{\frac{N}{2}-1} x(2r) W_{\frac{N}{2}}^{rK} - W_N^K \sum_{r=0}^{\frac{N}{2}-1} x(2r+1) W_{\frac{N}{2}}^{rK}$$

$$X\left(K+\frac{N}{2}\right) = G(k) - W_N{}^k H(k)$$

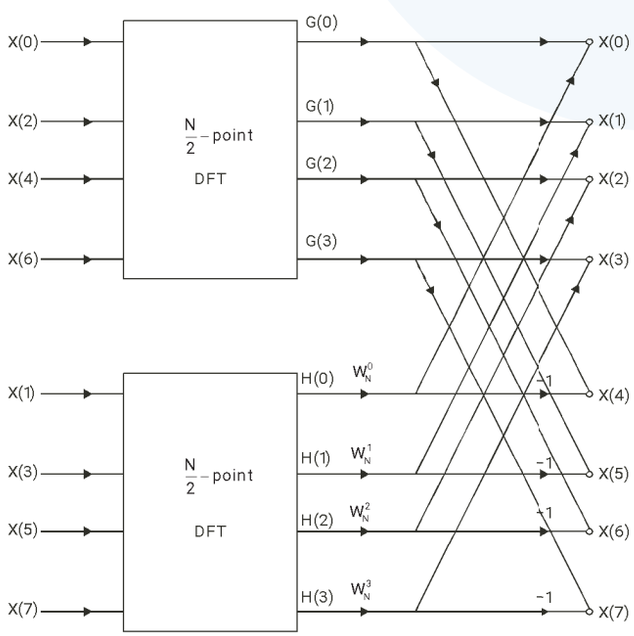


Fig. 3.1

Since G(K) and H(K) and both are periodic in K with period 4.

$$x(0) = G(0) + W_N^0 H(0)$$

$$x(1) = G(1) + W_N^1 H(1)$$

$$x(2) = G(2) + W_N^2 H(2)$$

$$x(3) = G(3) + W_N^3 H(3)$$

$$x(4) = G(0) - W_N^0 H(0)$$

$$x(5) = G(1) - W_N^1 H(1)$$

$$x(6) = G(2) - W_N^2 H(2)$$

$$x(7) = G(3) - W_N^3 H(3)$$

$$X(k) = \begin{cases} G(K) + W_N^K H(K), 0 \le K \le \frac{N}{2} - 1 \\ G(K) - W_N^K H(K+1), 0 \le K \le \frac{N}{2} - 1 \end{cases}$$

- The number of complex multiplications required for direct computation of the $\frac{N}{2}$ -point DFT G(k) = $\left(\frac{N}{2}\right)^2$
- The number of complex multiplications required for direct computation of the $\frac{N}{2}$ -point DFT H(k) = $\left(\frac{N}{2}\right)^2$
- There are $\frac{N}{2}$ additional complex multiplications required to compute W H(k).
- Total number of complex multiplications required to compute the N-point DFT with first decimation

$$= \left(\frac{N}{2}\right)^2 + \left(\frac{N}{2}\right)^2 + \frac{N}{2} = \frac{N^2}{2} + \frac{N}{2}$$

Note: In first decimation result, reduction of the number of multiplications from N^2 to $\frac{N^2}{2} + \frac{N}{2}$ which is a factor of 2 for N large.

The above process may be continued by expressing each of two point DFT's. G(K) and H(K) as a combination of two $\frac{N}{4}$ point DFTs, assuming that $\frac{N}{2}$ is even as it is always when N is equal to a power of 2. Each of the $\frac{N}{2}$ point DFTs in equation (i) is computed by breaking each such equation (i) into the $\frac{N}{4}$ point DFTs ,which is the combined to give the $\frac{N}{2}$ point DFTs. Thus, G(K) and H(K) in equation (i) would be computed as:

G(K)

$$= \sum_{r=0}^{\frac{N}{2}-1} g(r) W_{N/2}^{rk} = \sum_{l=0}^{\frac{N}{4}-1} g(2l) W_{N/2}^{2lk} + \sum_{l=0}^{\frac{N}{4}-1} g(2l+1) W_{N/2}^{(2l+1)k}$$

$$G(K) = \sum_{l=0}^{\frac{N}{4}-1} g(2l) W_{N/4}^{lk} + W_{N/2}^{k} \sum_{l=0}^{\frac{N}{4}-1} g(2l+1) W_{N/4}^{lk}$$

$$G(K) = P(K) + W_{N/2}^{K} Q(K)$$

P(K) = $\frac{N}{4}$-point DFT of even number of above sequence

Q(K) = $\frac{N}{4}$-point DFT of odd number of above sequence

$$G(K) = A(K) + W_{N/2}^{K} Q(K) \quad 0 \le K \quad \le \frac{N}{4} - 1$$

$$G\left(K + \frac{N}{4}\right) = P(K) - W_{N/2}^{K} Q(K) \quad 0 \le K \quad \le \frac{N}{4} - 1$$

Similarly,

$$H(K) = \sum_{l=0}^{\frac{N}{4}-1} h(2l) W_{N/4}^{lk} + W_{N/2}^{k} \sum_{l=0}^{\frac{N}{4}-1} h(2l+1) W_{N/4}^{lk}$$

$$H(K) = R(K) + W_{N/2}^{K} S(K) \quad 0 \le K \le \frac{N}{4} - 1$$

$$H\left(K + \frac{N}{4}\right) = R(K) - W_{N/2}^{K} S(K) \quad 0 \le K \le \frac{N}{4} - 1$$

In the above process of reducing an L-point DFT (L is power of 2) to an $\frac{L}{2}$ point DFTs, can be continued until we are left with 2-point DFTs, or there are L (= $\log_2$ N) stages, to be a evaluated. For N = 8 and $W_{N/2} = W_N^2$, we get

$$G(0) = P(0) + W_{N/2}^{0} Q(0) = P(0) + W_N^0 Q(0)$$

$$G(1) = P(1) + W_{N/2}^{1} Q(1) = P(1) + W_N^2 Q(1)$$

$$G(2) = P(0) - W_{N/2}^{0} Q(0) = P(0) - W_N^0 Q(0)$$

$$G(3) = P(1) - W_{N/2}^{1} Q(1) = P(1) - W_N^2 Q(1)$$

Similarly, for N = 8 and $W_{N/2} = W_N^2$, we get:

$$H(0) = R(0) + W_{N/2}^{0} S(0) = R(0) + W_N^0 S(0)$$

$$H(1) = R(1) + W_{N/2}^{1} S(1) = R(1) + W_N^2 S(1)$$

$$H(2) = R(0) - W_{N/2}^{0} S(0) = R(0) - W_N^0 S(0)$$

$$H(3) = R(1) - W_{N/2}^{1} S(1) = R(1) - W_N^2 S(1)$$

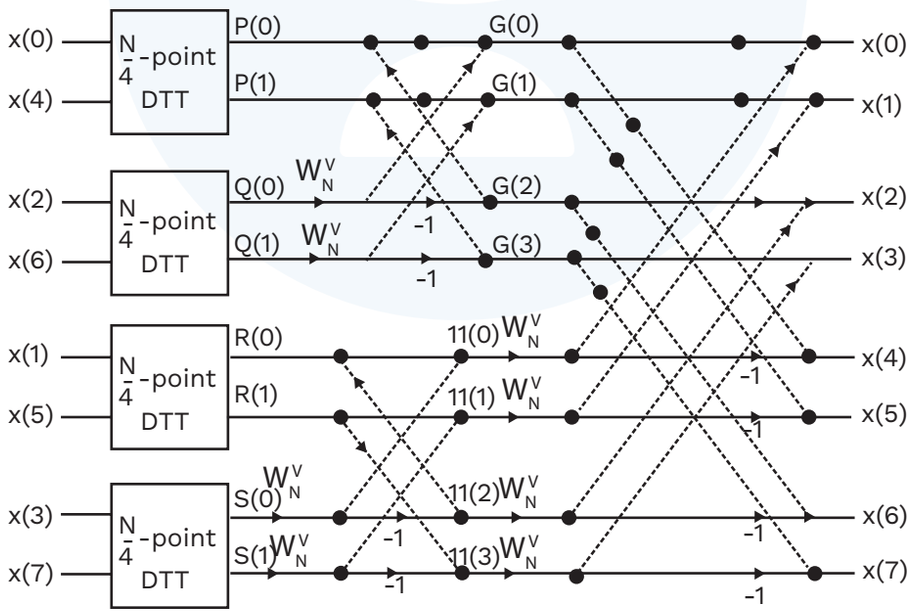


Fig. 3.2

- Each $\frac{N}{4}$-point direct computation of DFT's P(k), Q(k), R(k) & S(k) requires $\left(\frac{N}{4}\right)^2$ complex multiplications.
- Therefore, G(K) and H(K) requires $4\left(\frac{N}{4}\right)^2 + \frac{N}{4} + \frac{N}{4}$ = $\frac{N^2}{4} + \frac{N}{2}$ complex multiplications.

- Further, there are additional $\frac{N}{2}$ complex multiplication required to compute X(k) from G(k) and H(k).
- The total number of complex multiplications required to compute the N-point DFT with second decimation = $\frac{N^2}{4}+\frac{N}{2}+\frac{N}{2}=\frac{N^2}{4}+N$

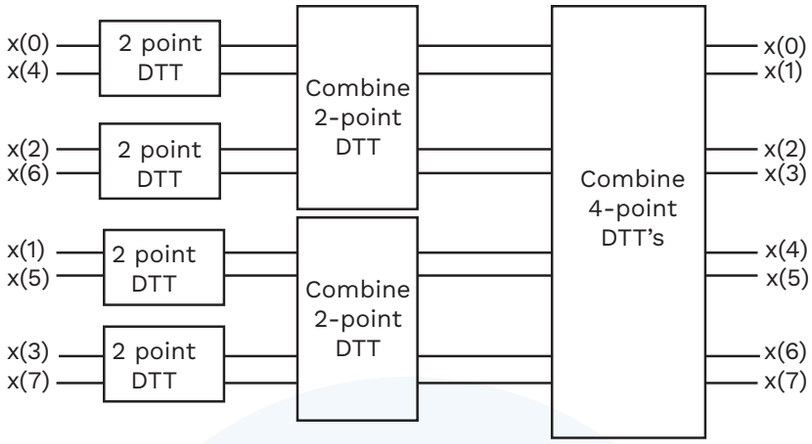


Fig. 3.3

Flow Graph of a 2-point DFT

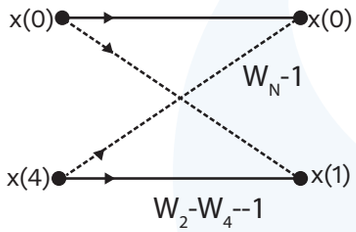


Fig. 3.4

- The number of complex multiplications and additions required are $N+2\left(\frac{N}{2}\right)^2$ or $N+\frac{N^2}{2}$ for the original N-point transform when decomposed into two $\frac{N}{2}$ point transforms.
- When $\frac{N}{2}$-point transforms are decomposed into $\frac{N}{4}$ point transforms, the factor of $\left(\frac{N}{2}\right)^2$ is replaced by $\frac{N}{2}+2\left(\frac{N}{4}\right)^2$. Hence, the overall computations required are $N+N+4\left(\frac{N}{4}\right)^2$ complex multiplications and additions.

Flow Graph of Complete Decimation in Time Decomposition of an 8-Point DFT Computation

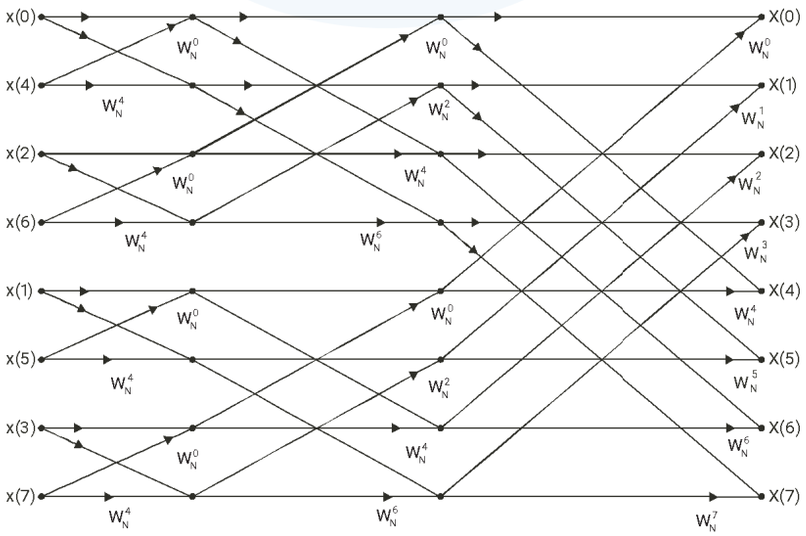


Fig. 3.5

Bit Reversal

In the flow graph of decimation in time decomposition of an 8-point DFT computation, the input data must be stored in a non-sequential order. In fact, the order in which the input data are stored is in bit reversed order. If $(n_2\ n_1\ n_0)$ is binary representation of the index of the sequence x(n), then the sequence value $X(n_2\ n_1\ n_0)$ is stored in the array position $X_0(n_0\ n_1\ n_2)$.

The sequence x(n) was first divided into even numbered samples and the odd numbered samples with the even numbered samples occurring in the top half and odd numbered samples occurring in the bottom half.

If least significant bit (n_0) is 0, the sequence value corresponds to even-numbered samples, and therefore, will appear in the top half, and if the least significant bit is 1, the sequence value corresponds to odd-numbered samples, and consequently, will appear in the bottom half of the array $X_0(l)$.

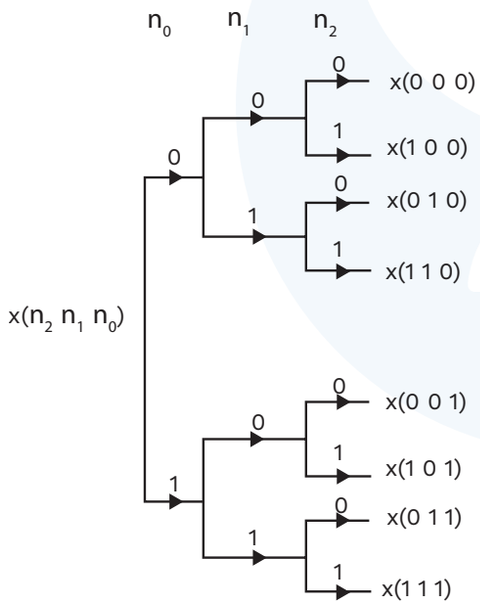


Fig. 3.6

Flow Graph of Basic Butterfly Computation in Decimation in Time (DIT) FFT Algorithm

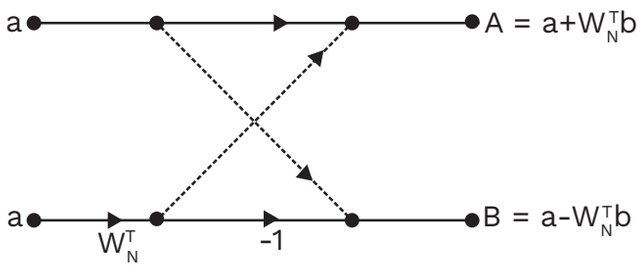


Fig. 3.7

Computational Advantage of DIT-FFT

The following observation can be made from the butterfly diagram of N = 8 point DIT-FFT algorithm.

- Number of stages, $L = \log_2 N = \log_2 8 = 3$
- Number of butterflies in each stage = $\frac{N}{2} = 4$
- Number of complex multiplications in each stage = $\frac{N}{2} = 4$
- Total number of complex multiplications = $\frac{N}{2}\log_2 N = 12$
- Number of complex additions in each stage = N = 8
- Total number of complex additions = $N \log_2 N = 24$
- Total number of real multiplications $= 4\left(\frac{N}{2}\log_2 N\right) = 2N\log_2 N = 48$
- Total number of real additions = $2\left(\frac{N}{2}\log_2 N\right) + 2(N\log_2 N) = 3\ N\ \log_2 N = 72$
- Input is in bit reversed order.
- Output is in normal order.

Flow Graph of 8-Point DFT Using Butterfly Computation

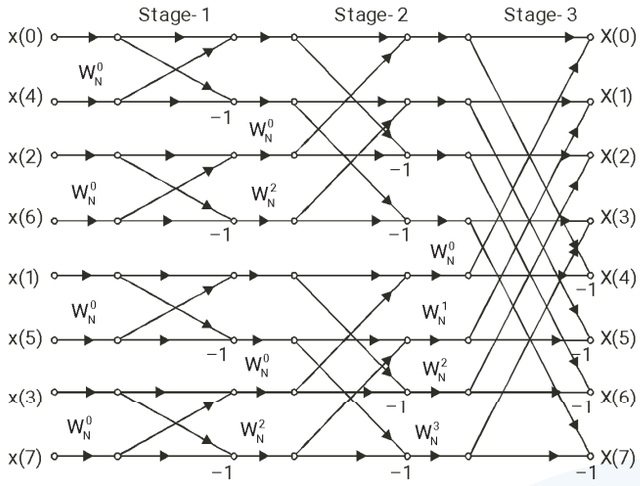


Fig. 3.8

- In general, each butterfly involves one complex multiplication and two complex additions.
- For $N = 2^L$, there are $\frac{N}{2}$ butterflies per stage of the computation process and $\log_2 N$ stages. Therefore, the total number of complex multiplications are $\frac{N}{2}\log_2 N$ and complex additions are $N\log_2 N$.
- Once a butterfly operation is performed on a pair of complex numbers (a, b) to produce (A, B), there is no need to save the input pair (a, b). Hence, we can store the result (A, B) in the same locations as (a, b). This is called in place computation.
- For an N-point DFT, we need N complex registers or 2N real registers (or memory locations) for the first stage. In addition, two real registers (or memory location) are required to store one twiddle factor for each butterfly. As there are $\frac{N}{2}$ butterflies in each stage, so $\frac{N}{2} \times 2 = N$ additional memory locations are required for the storage of twiddle factors in each stage. Thus, the total memory locations = 2N + N + 3N. The same memory locations will be used for the subsequent stages.

Solved Examples

1. Assume that a complex multiply takes 1μs and that the amount of time to compute a DFT is determined by the amount of time it takes to perform all the multiplications.

a. How much time does it take to compute a 1024-point D.F.T directly?

b. How much time is required if an FFT is used?

c. Repeat parts a and b for a 4096-point DFT?

Solution:

a. N-point DFT directly requires N^2 complex multiplications. If it takes 1μs per complex multiply, the direct evaluation of a 1024 point DFT requires.

$t_{DFT} = (1024)^2 \times 10^{-6} = 1.05$ s

b. With a radix-2 FFT, the number of complex multiplications are approximately $\frac{N}{2}\log_2 N$.

For N = 1024

Number of complex multiplications = $512 \log_2 1024 = 5120$

The amount of time to compute a 1024-point DFT using FFT is

$t_{FFT} = 5120 \times 10^{-6} = 5.12\mu s$

c. If the length of DFT is increased by a factor of 4 to N = 4096, the number of multiplications necessary to compute DFT directly increases by a factor of 16. Therefore: the time required to evaluate the DFT directly is:

$t_{DFT} = 16 \times 1.05$

$t_{DFT} = 16.78$ S

With a radix-2 FFT

Number of complex multiplications = $\frac{N}{2}\log_2 N$

$= 2048 \log_2 4096$

$= 24576$

Amount of time to compute a 4096-point DFT using FFT is:

$t_{FFT} = 24576 \times 10^{-6}$

$t_{FFT} = 24.576$ ms

2. Given x(n) = {1,2,3,4,4,3,2,1}. Find X(k) using decimation in time FFT algorithm

Solution:

$W_N^K = e^{-j\left(\frac{2\pi}{N}\right)K}$, N = 8

$W_8^0 = e^{-j\left(\frac{2\pi}{8}\right)0} = 1$

$W_8^1 = e^{-j\left(\frac{2\pi}{8}\right)1} = e^{-j\frac{\pi}{4}} = \cos\frac{\pi}{4} - j\sin\frac{\pi}{4} = 0.707 - j0.707$

$W_8^2 = e^{-j\left(\frac{2\pi}{8}\right)2} = e^{-j\frac{\pi}{2}} = -j$

$W_8^3 = e^{-j\left(\frac{2\pi}{8}\right)3} = -0.707 - j0.707$

Using DFT FFT algorithm, we can find Z(k) from the given sequence x(n) as shown in the figure given below.

Therefore,

X(k) = (20, −5.828 − j2.414, 0, 0.172 − j 0.414, 0, − 0.172 + j 0.414, 0, − 5.828 + j2.414)

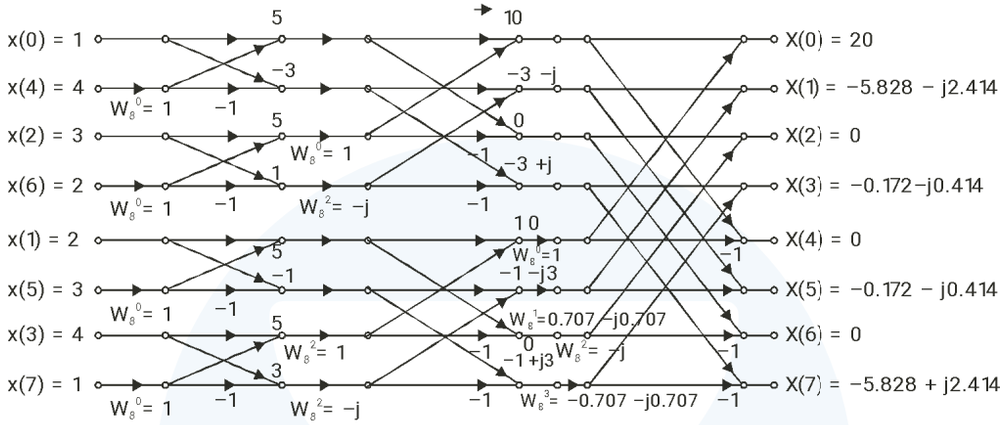


Fig. 3.9

3. Given x(n) = (0, 1, 2, 3, 4, 5, 6, 7), find X(k) using D.I.T. FFT algorithm.

Solution:

Given N = 8

$W_N^K = e^{-j\left(\frac{2\pi}{N}\right)k}$

Hence, $W_8^0 = 1$

$W_8^1 = 0.707 - j0.707$

$W_8^2 = -j$

$W_8^3 = -0.707 - j0.707$

Using D.T.T. FFT algorithm, we can find X(k) from the given sequence x(n) as shown in figure given below.

Therefore, X(k) = (28, −4 + j9.656, −4 + 4j, −4 + j1.656, −4, −4 −j1.656, −4 −4j, −4 −j9.656)

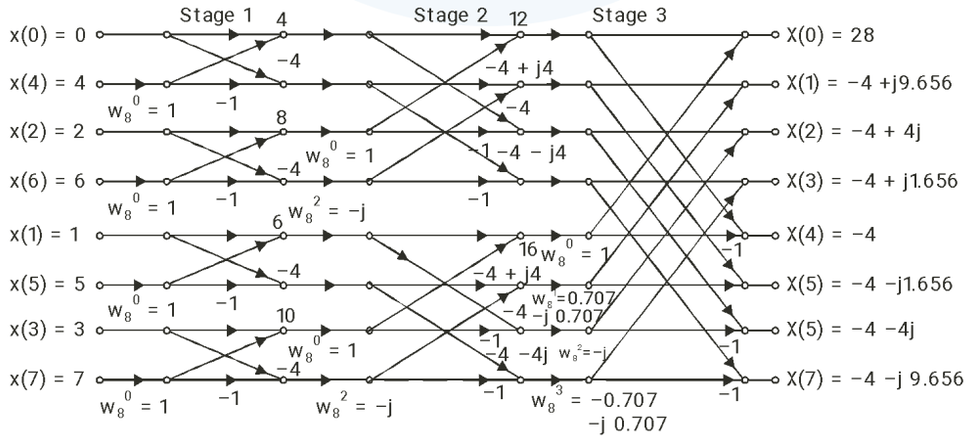


Fig. 3.10

4. Given $x(n) = 2^n$ and N = 8, find X(k) using D.I.T. FFT algorithm.

Solution:

Given $x(n) = 2^n$ and N = 8

x(0) = 1, x(1) = 2, x(2) = 4, x(3) = 8

x(4) = 16, x(5) = 32, x(6) = 64, x(7) = 128

x(n) = {1, 2, 4, 8, 16, 32, 64, 128}

$W_8^0 = 1$

$W_8^1 = 0.707 - j\ 0.707$

$W_8^2 = -j$

$W_8^3\ x = -0.707 - j0.707$

Using D.T.T. FFT algorithm, we can find X(k) from the given sequence x(n) as shown in figure given below.

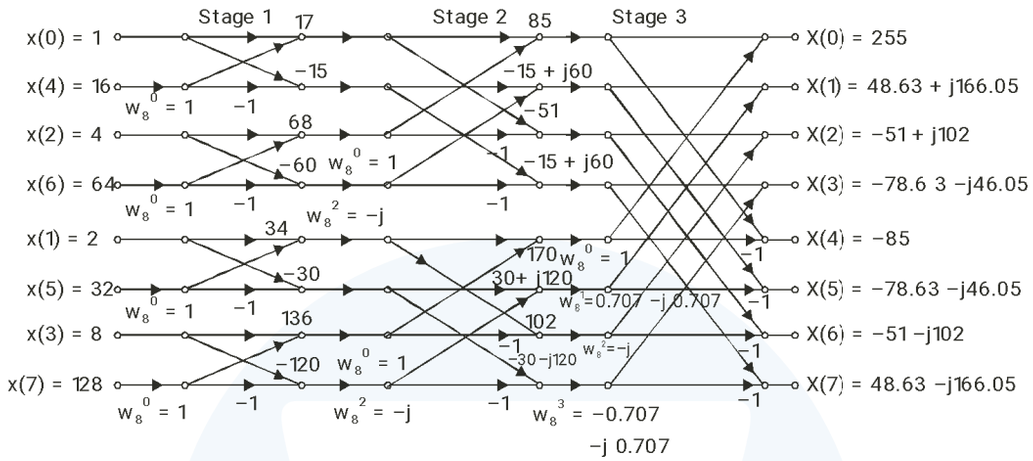


Fig. 3.11

X(k) = (255, 48.63 + j 166.05, −51 + j 102, −78.63 + j 46.05, −85, −78.63 −j46.05, −51 − j102, 48.63 − j166.05)

Decimation in Frequency Algorithm (DIF-FFT)

In decimation in frequency, we decimate the N-point DFT sequence X(k) into smaller subsequences.

$$X(K) = \sum_{n=0}^{N-1} x(n) W_N^{kn}$$

$$= \sum_{n=0}^{\frac{N}{2}-1} x(n) W_N^{kn} + \sum_{n=\frac{N}{2}}^{N-1} x(n) W_N^{kn}$$

$$= \sum_{n=0}^{\frac{N}{2}-1} x(n) W_N^{kn} + \sum_{n=0}^{N-1} x\left(n+\frac{N}{2}\right) W_N^{k\left(n+\frac{N}{2}\right)}$$

$$= \sum_{n=0}^{\frac{N}{2}-1} x(n) W_N^{kn} + W_N^{k\frac{N}{2}} \sum_{n=0}^{\frac{N}{2}-1} x\left(n+\frac{N}{2}\right) W_N^{kn}$$

$$W_N^{k\frac{N}{2}} = e^{-j\frac{2\pi}{N}\cdot\frac{KN}{2}} = e^{-j\pi K} = \left(e^{-j\pi}\right)^K = (-1)^K$$

$$X(K) = \sum_{n=0}^{\frac{N}{2}-1} x(n) W_N^{Kn} + (-1)^K \sum_{n=0}^{\frac{N}{2}-1} x\left(n+\frac{N}{2}\right) W_N^{Kn}$$

$$X(K) = \sum_{n=0}^{\frac{N}{2}-1} \left[x(n) + (-1)^K x\left(n+\frac{N}{2}\right) \right] W_N^{Kn}, 0 \le x \le 1$$

Decimating X(K) into even-indexed and odd-indexed samples:

X(2K)

$$= \sum_{n=0}^{\frac{N}{2}-1} \left[x(n) + (-1)^{2K} x\left(n+\frac{N}{2}\right) \right] W_N^{2Kn}, 0 \le k \le \frac{N}{2} - 1 \rightarrow$$

even indexed

X(2K +1) =

$$\sum_{n=0}^{\frac{N}{2}-1} \left[x(n) + (-1)^{2K+1} x\left(n+\frac{N}{2}\right) \right] W_N^{(2K+1)n}, 0 \le K \le \frac{N}{2} - 1 \rightarrow$$

odd indexed

$(-1)^{2K} = 1$, $(-1)^{2K+1} = -1$, $W_N^{2Kn} = W_{N/2}^{Kn}$

$$X(2K) = \sum_{n=0}^{\frac{N}{2}-1} \left[x(n) + x\left(n+\frac{N}{2}\right) \right] W_{N/2}^{Kn}, 0 \le K \le \frac{N}{2} - 1$$

X(2K + 1)

$$= \sum_{n=0}^{\frac{N}{2}-1} \left\{ \left[x(n) - x\left(n+\frac{N}{2}\right) \right] W_N^n \right\} W_{N/2}^{Kn}, 0 \le K \le \frac{N}{2} - 1$$

Let $g(n) = x(n) + x\left(n + \frac{N}{2}\right), 0 \le n \le \frac{N}{2} - 1$

$h(n) = \left[x(n) - x\left(n + \frac{N}{2}\right)\right] W_N^n, 0 \le n \le \frac{N}{2} - 1$

$G(K) = X(2K) = \sum_{n=0}^{\frac{N}{2}-1} g(n) W_{N/2}^{kn}, 0 \le k \le \frac{N}{2} - 1$

$H(K) = X(2K + 1) = \sum_{n=0}^{\frac{N}{2}-1} h(n) W_{N/2}^{kn}, 0 \le x \le \frac{N}{2} - 1$

For N = 8

$g(0) = x(0) + x(4)$

$g(1) = x(1) + x(5)$

$g(2) = x(2) + x(6)$

$g(3) = x(3) + x(7)$

$h(0) = \left[x(0) - x(4)\right] W_N^0$

$h(1) = \left[x(1) - x(5)\right] W_N^1$

$h(2) = \left[x(2) - x(6)\right] W_N^2$

$h(3) = \left[x(3) - x(7)\right] W_N^3$

$G(0) = x(0)$

$G(1) = x(2)$

$G(2) = x(4)$

$G(3) = X(6)$

$H(0) = X(1)$

$H(1) = X(3)$

$H(2) = X(5)$

$H(3) = X(7)$

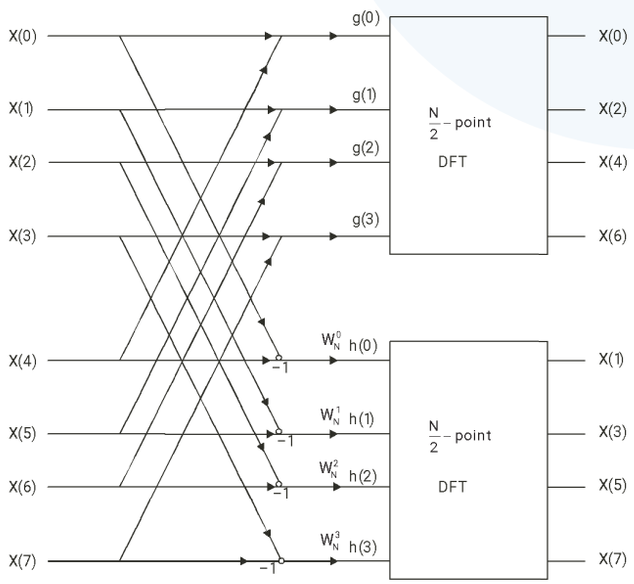


Fig. 3.12

Now, decimate the $\frac{N}{2}$-point DFT G(K) = X(2K) into two $\frac{N}{4}$-point DFTs.

$$G(K) = \sum_{n=0}^{\frac{N}{2}-1} g(n) W_{N/2}^{Kn} = \sum_{n=0}^{\frac{N}{4}-1} g(n) W_{N/2}^{Kn} + \sum_{n=\frac{N}{4}}^{\frac{N}{2}-1} g(n) W_{N/2}^{Kn}$$

$$= \sum_{n=0}^{\frac{N}{4}-1} g(n) W_{N/2}^{Kn} + \sum_{n=0}^{\frac{N}{4}-1} g\left(n + \frac{N}{4}\right) W_{N/2}^{K\left(n+\frac{N}{4}\right)}$$

$$= \sum_{n=0}^{\frac{N}{4}-1} g(n) W_{N/2}^{Kn} + W_{N/2}^{K\frac{N}{4}} \sum_{n=0}^{\frac{N}{4}-1} \left(n + \frac{N}{4}\right) W_{N/2}^{Kn}$$

$$W_{N/2}^{KN/4} = e^{-j\frac{2\pi K}{N/2}\left(\frac{N}{4}\right)} = e^{-j\frac{4\pi K}{N}\cdot\frac{N}{4}} = e^{-j\pi k} = (-1)^K$$

$$G(K) = \sum_{n=0}^{\frac{N}{4}-1} g(n) W_{N/2}^{Kn} + (-1)^K \sum_{n=0}^{\frac{N}{4}-1} g\left(n + \frac{N}{4}\right) W_{N/2}^{Kn}$$

$$G(K) = \sum_{n=0}^{\frac{N}{4}-1} \left[g(n) + (-1)^K g\left(n + \frac{n}{4}\right)\right] W_{N/2}^{Kn}$$

Decimating G(K) into even-indexed and odd-indexed samples.

$G(2K) = X(4K)$

$$= \sum_{n=0}^{\frac{N}{4}-1} \left[g(n) + (-1)^{2K} g\left(n + \frac{N}{4}\right)\right] W_{N/2}^{2Kn}, 0 \le K \le \frac{N}{4} - 1$$

$G(2K + 1) = X(4K + 2) =$

$$\sum_{n=0}^{\frac{N}{4}-1} \left[g(n) + (-1)^{2K+1} g\left(n + \frac{N}{4}\right)\right] W_{N/2}^{(2K+1)n}, 0 \le K \le \frac{N}{4} - 1$$

$(-1)^{2K} = 1$, $(-1)^{2K+1} = -1$, $W_{N/2}^{2Kn} = W_{N/4}^{kn}$

$$G(2K) = \sum_{n=0}^{\frac{N}{4}-1} \left[g(n) + g\left(n + \frac{N}{4}\right)\right] W_{N/4}^{Kn}, 0 \le K \le \frac{N}{4} - 1$$

$G(2K + 1)$

$$= \sum_{n=0}^{\frac{N}{4}-1} \left\{\left[g(n) - g\left(n + \frac{N}{4}\right)\right] W_{N/2}^{n}\right\} W_{N/4}^{Kn}, 0 \le K \le \frac{N}{4} - 1$$

$$a(n) = g(n) + g\left(n + \frac{N}{4}\right), 0 \le n \le \frac{N}{4} - 1$$

$$b(n) = \left[g(n) - g\left(n + \frac{N}{4}\right)\right] W_{N/2}^{n}, 0 \le n \le \frac{N}{4} - 1$$

$$= \left[g(n) - g\left(n + \frac{N}{4}\right)\right] W_N^{2n} \left\{W_{N/2}^n = W_N^{2n}\right\}$$

$$A(K) = G(2K) = X(4k) = \sum_{n=0}^{\frac{N}{4}-1} a(n) W_{N/4}^{Kn}, 0 \le n \le \frac{N}{4} - 1$$

$$B(K) = G(2K + 1) = X(4K + 2)$$

$$= \sum_{n=0}^{\frac{N}{4}-1} b(n) W_{N/4}^{Kn}, 0 \le k \le \frac{N}{4} - 1$$

a(0) = g(0) + g(2)	A(0) = G(0) = X(0)
a(1) = g(1) + g(3)	A(1) = G(2) = X(4)
a(0) = [g(0) – g(2)] W_N^0	B(0) = G(1) = X(2)
b(1) = [g(1) –g(3)] W_N^2	B(1) = G(3) = X(6)

Similarly, decimating H(K) even-indexed and odd-indexed samples.

$$C(K) = H(2K) = X(4K+1) = \sum_{n=0}^{\frac{N}{4}-1} c(n) W_{N/4}^{kn}, 0 \le K \le \frac{N}{4} - 1$$

$$D(K) = H(2K+1) = X(4K+3) = \sum_{n=0}^{\frac{N}{4}-1} d(n) W_{N/4}^{kn}, 0 \le K \le \frac{N}{4} - 1$$

Where,

$$c(n) = h(n) + h\left(n + \frac{N}{4}\right), 0 \le n \le \frac{N}{4} - 1$$

$$d(n) = \left[h(n) - h\left(n + \frac{N}{4}\right)\right] W_{N/2}^{n}, 0 \le n \le \frac{N}{4} - 1$$

$$d(n) = \left[h(n) - h\left(n + \frac{N}{4}\right)\right] W_N^{2n}$$

c(0) = h(0) + h(2)	C(1) = H(0) = X(1)
c(0) = h(0) + h(2)	C(0) = H(0) = X(1)
c(1) = h(1) + h(3)	C(1) = H(2) = X(5)
d(0) = $\left[h(0) - h(2)\right] W_N^0$	D(0) = H(1) = X(3)
d(1) = $\left[h(1) - h(3)\right] W_N^2$	D(1) = H(3) = X(7)

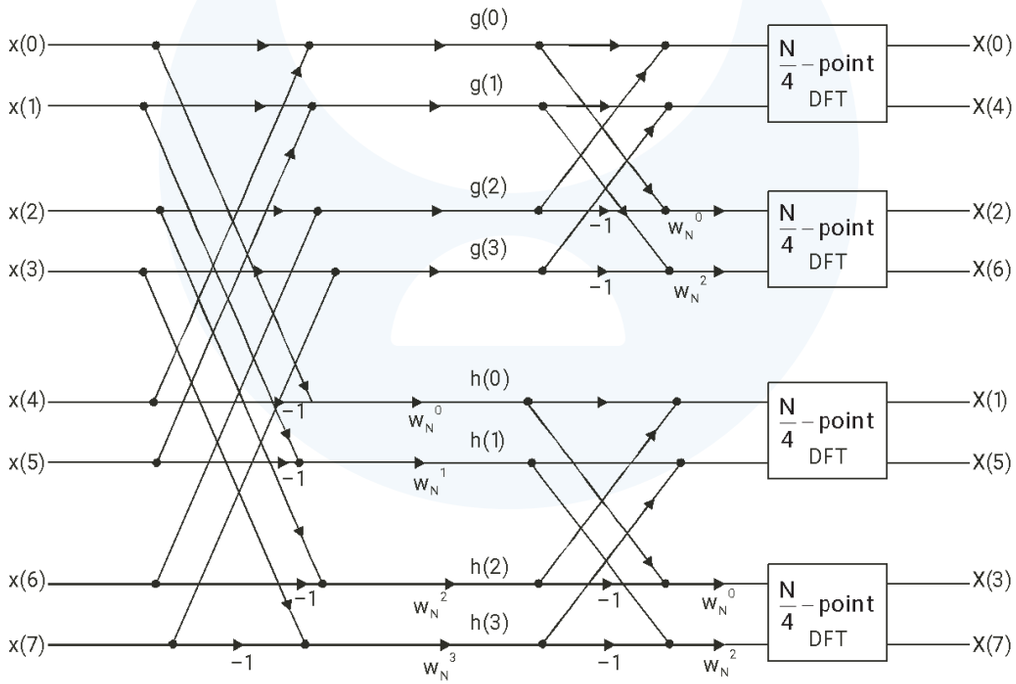


Fig. 3.13

This decimation process can continue until the smallest DFTs are 2-point DFTs.

Flow graph of basic butterfly computation is decimation in frequency FFT algorithm

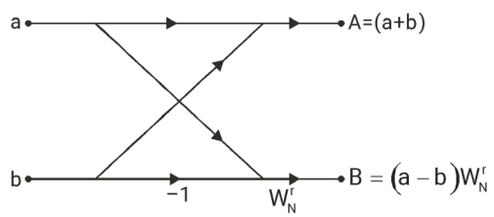


Fig. 3.14

Signal Flow Graph (Butterfly Diagram of DIF–FFT algorithm for N = 8)

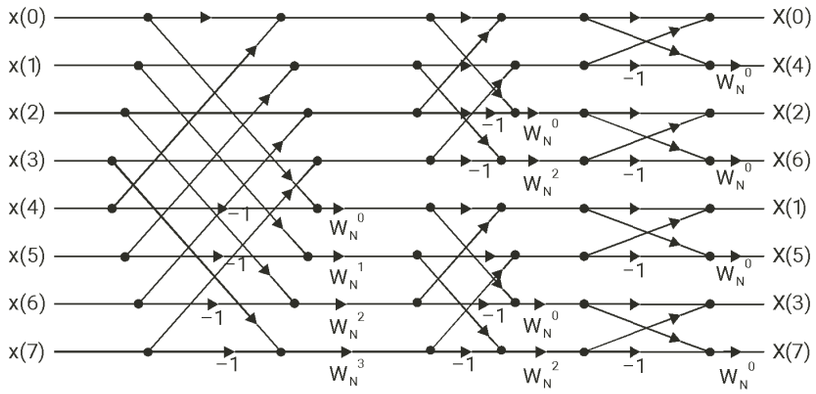


Fig. 3.15

Computational Advantage of DIF–FFT

The following observations can be made from the butterfly diagram of an N = 8 DIF-FFT algorithm.

- The input is in normal order.
- The output is in bit reversal.
- Computation are performed in place.
- Number of stages $(m) = \log_2^N = \log_2^8 = 3$.
- Number of butterflies in each stage = $\frac{N}{2} = 4$
- Number of complex multiplications in each stage = $\frac{N}{2} = 4$
- Total number of complex multiplications = $\frac{N}{2}\log_2 N = 12$
- Number of complex additions in each stage = N = 8
- Total number of complex additions = $N\log_2 N = 24$
- Total number of real multiplication $= 4\left(\frac{N}{2}\log_2 N\right) = 2N\log_2 N = 48$
- Total number of real additions = $2\left(\frac{N}{2}\log_2 N\right) + 2\left(N\log_2 N\right) = 3N\log_2 N = 72$
- Total memory location = 3N = 24

Applications of FFT Algorithms

- Efficient computation of the DFT of two real sequences.
- Efficient computation of the DFT of a 2N point real sequence.
- Use of the FFT algorithm in linear filtering and correlation

Solved Examples

5. Compute the 2-point DFT X(k) using DIT-FFT algorithm of a length-2 sequence x(n). Draw the butterfly diagram.

Solution:

Given that x(n) = {x(0), x(1)}. By definition, N = 2-point DFT is given by:

$$X(k) = \sum_{n=0}^{1} x(n) W_N^{nk}, 0 \le k \le 1$$

$$X(0) = x(0) + x(1)$$

$$X(1) = x(0) + W_2^1 x(1)$$

$$= x(0) + e^{-j\frac{2\pi}{2}} x(1)$$

$$= x(0) + e^{-j\pi} x(1) = x(0) - x(1)$$

Butterfly diagram is shown in the following figure.

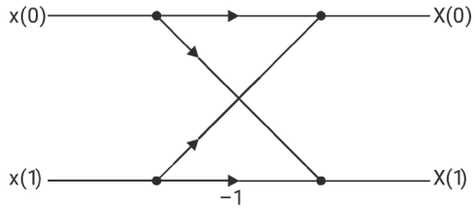


Fig. 3.16 Butterfly Diagram for N = 2

6. Compute the 4-point DFT X(k) using DIT FFT algorithm of a length-4 sequence x(n), $0 \le n \le 3$. Draw the butterfly diagram.

Solution:

Given that x(n) = {x(0), x(1), x(2), x(3)}. Decimating the sequence x(n) into even-indexed and odd-indexed values, we obtain:

g(n) = x(2) = {x(0), x(2)}

h(n) = x(2n + 1) = {x(1), x(3)}

By definition, 4-point DFT is given by:

$$X(k) = \sum_{n=0}^{3} x(n) W_4^{nk}$$

$$= \sum_{n=0}^{1} x(2n) W_4^{2nk} + \sum_{n=0}^{1} x(2n+1) W_4^{(2n+1)k}$$

$$= \sum_{n=0}^{1} g(n) W_2^{nk} + W_4^k \sum_{n=0}^{1} h(n) W_2^{nk}$$

$$X(k) = G(k) + W_4^k H(k), 0 \le k \le 3$$

Where, G(k) and H(k) are 2-point DFT Both are periodic with period 2, that is G(k) = G(k+2) and H(k) = H(k+2), for $0 \le k \le 1$. Since $W_4^{k+2} = -W_4^k$, this equation can be written as:

$$X(k) = G(k) + W_4^k H(k), 0 \le k \le 1$$

$$X(k+2) = G(k) - W_4^k H(k), 0 \le k \le 1$$

Thus,

$$X(0) = G(0) + W_4^0 H(0)$$

$$X(2) = G(0) - W_4^0 H(0)$$

$$X(1) = G(1) + W_4^1 H(1)$$

$$X(3) = G(1) - W_4^1 H(1)$$

Now, consider the 2-point DFT G(k) and H(k).

$$G(k) = \sum_{n=0}^{1} g(n) W_2^{nk}, 0 \le k \le 1$$

G(0) = g(0) + g(1) = x(0) + x(2)

$$G(1) = g(0) + W_2^1 g(1)$$

$$= g(0) - W_4^0 g(1) = x(0) - W_4^0 X(2)$$

Similarly,

$$H(k) = \sum_{n=0}^{1} h(n) W_2^{nk}, 0 \le k \le 1$$

H(0) = h(0) + h(1) = x(1) + x(3)

$$H(1) = h(0) + W_2^1 h(1)$$

$$= h(0) - W_4^0 h(1) = x(1) - W_4^0 x(3)$$

Butterfly diagram for 4-point DIT-FFT algorithm shown in below Fig.

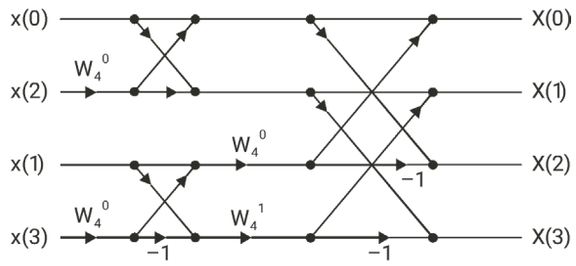


Fig. 3.17 Butterfly Diagram for 4-point DIT-FFT Algorithm

7. Given $x(n) = \left\{\underset{\uparrow}{0}, 1, 2, 3, 4, 5, 6, 7\right\}$, find 8-point DFT X(k) using the D.I.T.-FFT algorithm. Show all intermediate results.

Solution:

Given that N = 8. We know that $W_N^k = e^{-j\frac{2\pi}{N}k}$. Therefore,

$$W_8^0 = e^{-j\frac{2\pi}{N}0} = 1$$

$$W_8^1 = e^{-j\frac{2\pi}{8}} = e^{-j\frac{\pi}{4}}$$

$$= \cos\frac{\pi}{4} - j\sin\frac{\pi}{4} = 0.707 - j0.707$$

$$W_8^2 = e^{-j\frac{2\pi}{8}2} = e^{-j\frac{\pi}{2}} = \cos\frac{\pi}{2} - j\sin\frac{\pi}{2} = -j$$

$$W_8^3 = e^{-j\frac{2\pi}{8}3} = e^{-j\frac{3\pi}{4}}$$

$$= \cos\frac{3\pi}{4} - j\sin\frac{3\pi}{4} = -0.707 - j0.707$$

The butterfly diagram of an 8-point DIT-FFT algorithm is shown in the figure given below.

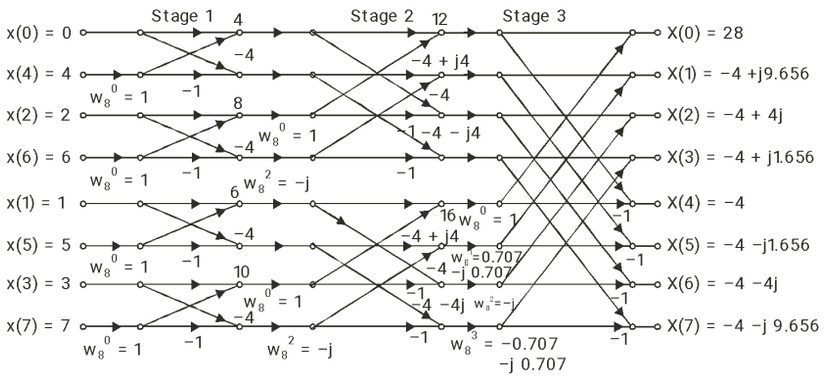


Fig. 3.18

$$X(k) = \left\{\begin{matrix}\underset{\uparrow}{28}, -4 + j9.656, -4 + j4 - 4 + j3.656, \\ -4, -4 - j1.656, -4 - j4, -4 - j9.656\end{matrix}\right\}$$

8. Compute the 8-point DFT of time sequence $x(n) = \cos\left(\frac{n\pi}{2}\right)$ using the DIT-FFT algorithm. Show all intermediate results.

or

Given $x(n) = \left\{\underset{\uparrow}{1}, 0, -1, 0, 1, 0, -1, 0\right\}$, find X(k) using the DIT-FFT algorithm. Show all intermediate results.

Solution:

Given that N = 8 and $x(n) = \cos\left(\frac{\pi n}{2}\right), 0 \le n \le 7$

$$x(n) = \cos\left(\frac{n\pi}{2}\right) = \left\{\underset{\uparrow}{1}, 0, -1, 0, 1, 0, -1, 0\right\}$$

We know that $W_N^k = e^{-j\frac{2\pi}{N}k}$, therefore,

$W_8^0 = j, W_8^1 = 0.707 - j0.707$

$W_8^2 = -j, W_8^3 = -0.707 - j0.707$

The butterfly diagram of an 8-point DIT-FFT algorithm is shown in figure given below.

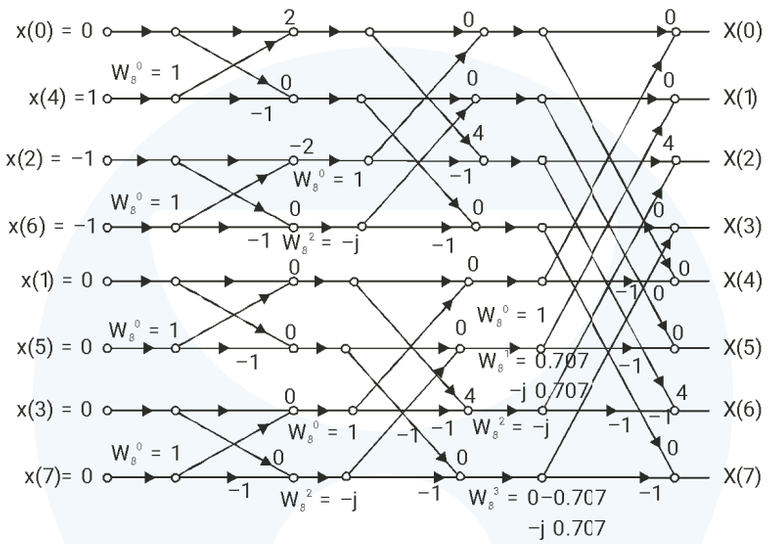


Fig. 3.19

$$X(k) = \left\{\underset{\uparrow}{0}, 0, 4, 0, 0, 0, 4, 0\right\}$$

9. Sample the signal $x(t) = 1 + \cos 2\pi t$ at its Nyquist rate and obtain its discrete sequence. Determine 4-point DFT using DIT-FFT algorithm.

Solution:

Given that $x(t) = 1 + \cos 2\pi t$. The maximum frequency contained in x(t) is $f_{max} = 1$ Hz. Therefore, the Nyquist frequency is $f_S = 2f_{max} = 2$ and $T_s = \frac{1}{S} = \frac{1}{2}$. Sampling x(t) at t = nTs, we obtain

$$x(t)\big|_{t=nT_s} = x(nT_s) = x(n) = 1 + \cos(2\pi n T_s)$$

$$= 1 + \cos(n\pi), \quad 0 \le n \le 3$$

$$x(n) = \left\{\underset{\uparrow}{2}, 0, 2, 0\right\}$$

Given that N = 4. We know that $W_N^k = e^{-j\frac{2n}{N}k}$. Therefore,

$$W_4^0 = e^{-j\frac{2\pi}{4}0} = 1,$$

$$W_4^1 = e^{-j\frac{2\pi}{4}} = e^{-j\frac{\pi}{2}} = \cos\frac{\pi}{2} - j\sin\frac{\pi}{2} = -j$$

The butterfly diagram of a 4-point DIT-FFT algorithm is shown in figure given below.

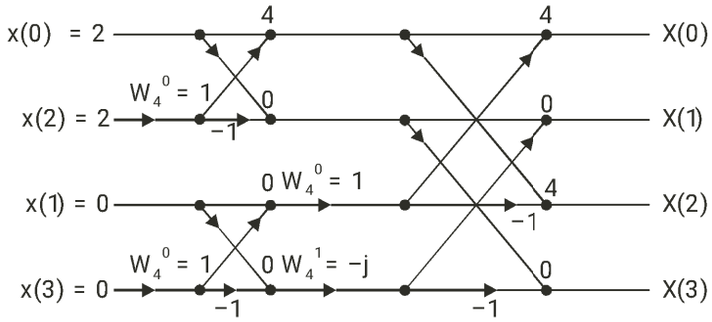


Fig. 3.20

$X(k) = \{\underset{\uparrow}{4}, 0, 4, 0\}$

Comparison between DIT/DIF-FFT and Direct Computation

Feature	N-point DFT	N-point DIT / DIF-FFT
Algorithm	Solution of N equations in N unknowns	$\frac{N}{2}$ butterflies per stage, $\log_2 N$ Stages, total butterflies = $\frac{N}{2}\log_2 N$
Complex multiplications	N per equation	1 per butterfly
Complex additions	N−1 per equation	2 per butterfly
Total multiplications	N^2	$\frac{N}{2}\log_2 N$
Total additions	N (N − 1)	N $\log_2$ N

Table 3.1

Comparison between DIT and DIF Algorithms

DIT-FFT	DIF-FFT
Differences	
• The time-domain sequence is decimated • The input sequence is in bit-reversed order • while the output is in normal order • In butterfly diagram, the complex multiplication takes place before the add-subtract operation.	• The frequency-domain sequence is decimated • The input sequence is in normal order while the output is in normal order • In butterfly diagram, the complex multiplication takes place after the add-subtract operation.
Simltarities	
• Number of stages = m, and $N = 2^m$ • Number of butterflies per stages = $\frac{N}{2}$ • Number of complex multiplications = $\frac{N}{2}\log_2 N$ • Number of complex additions = N $\log_2$ N • Computations are in place	• Number of stages = m, and $N = 2^m$ • Number of butterflies per stages = $\frac{N}{2}$ • Number of complex multiplications = $\frac{N}{2}\log_2 N$ • Number of complex additions = N $\log_2$ N • Computations are in place

Table 3.2

10. Compute the 4-point DFT X(k) using DIF-FFT algorithm of a length-4 sequence x(n), $0 \le n \le 3$. Draw the butterfly diagram.

Solution:

Given that N = 4 and x(n) = {x(0), x(1), x(2), x(3)}. Dividing the input sequence x(n) into two halves leads to:

$$X(k) = \sum_{n=0}^{3} x(n) W_4^{kn}$$

$$= \sum_{n=0}^{1} x(n) W_4^{kn} + \sum_{n=2}^{3} z(n) W_4^{kn}$$

$$= \sum_{n=0}^{1} x(n) W_4^{kn} + \sum_{n=2}^{3} x(n+2) W_4^{k(n+2)}$$

$$X(k) = \sum_{n=0}^{1} x(n) W_4^{kn} + W_4^{2k} \sum_{n=2}^{3} x(n+2) W_4^{kn}$$

Since $W_4^{2k} = e^{\frac{-j2\pi 2k}{4}} = e^{-j\pi k} = \left(e^{-j\pi}\right)^k = (-1)^k$, we have

$$X(k) = \sum_{n=0}^{1} x(n) W_4^{kn} + (-1)^k \sum_{n=0}^{1} x(n+2) W_4^{kn}$$

$$X(k) = \sum_{n=0}^{1} \left[x(n) + (-1)^k x(n+2)\right] W_4^{kn}, 0 \le k \le 3$$

Then, decimate X(k) into even-indexed and odd-indexed samples. The term in the even-indexed sequence may be designated to X(2k), $0 \le k \le 1$, while those in the odd-indexed sequence becomes X(2k+1), $0 \le k \le 1$. Thus, this equation can be written as:

$$X(2k) = \sum_{n=0}^{1} \left[x(n) + (-1)^{2k} z(n+2)\right] W_4^{2kn}, 0 \le k \le 1$$

$$X(2k+1) = \sum_{n=0}^{1} \left[x(n) + (-1)^{2k+1} x(n+2)\right]$$

$$W_4^{(2k+1)n}, 0 \le k \le 1$$

Since, $(-1)^{2k} = 1$, $(-1)^{2k+1} = -1$, and $W_4^{2kn} = W_2^{kn}$

$$X(2k) = \sum_{n=0}^{1} \left\{x(n) + x(n+2)\right\} W_2^{kn}, \ 0 \le k \le 1$$

$$X(2k+1) = \sum_{n=0}^{1} \left\{\left[x(n) - x(n+2)\right] W_4^k\right\} W_2^{kn}, \ 0 \le k \le 1$$

Let us define two 2-point sequence g(n) and h(n) as:

$g(n) = x(n) + x(n+2), \ 0 \le n \le 1$

$h(n) = \{x(n) - x(n+2)\} \ W_4^n, \ 0 \le n \le 1$

Thus, $g(0) = x(0) + x(2)$, $h(0) = \left[x(0) - x(2)\right] W_4^0$

$g(1) = x(1) + x(3)$, $h(1) = \left[x(1) - x(3)\right] W_4^1$

Using g(n) and h(n) we can write:

$$X(2k) = G(k) = \sum_{n=0}^{1} g(n) W_2^{kn}, 0 \le k \le 1$$

$$X(2k+1) = H(k) = \sum_{n=0}^{1} h(n) W_2^{kn}, 0 \le k \le 1$$

and we obtain:

$X(0) = G(0) = g(0) + g(1)$

$X(2) = G(1) = g(0) + W_2^1 g(1) = g(0) - W_4^0 g(1)$

$X(1) = H(0) = h(0) + h(1)$

$X(3) = H(1) = h(0) + W_2^1 h(1) = h(0) - W_4^0 h(1)$

Butterfly diagram for 4-point DIF-FFT algorithm shown in the figure below.

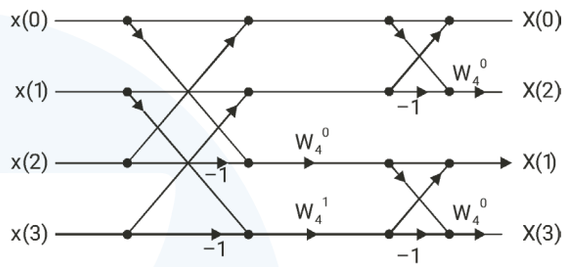


Fig. 3.21 Butterfly Diagram for 4-point DIF-FFT Algorithm

11. Find the DFT of the sequence $x(n) = \left\{\underset{\uparrow}{4}, 3, 2, 1\right\}$ using DIF-FFT algorithm.

Solution:

Given that N = 4, we know that $W_N^k = e^{\frac{-j2\pi}{N}k}$

Therefore, $W_4^0 = 1$

$$W_4^1 = e^{-j\frac{2\pi}{4}} = e^{-j\frac{\pi}{2}} = \cos\frac{\pi}{2} - j\sin\frac{\pi}{2} = -j$$

The butterfly diagram of a 4-point DIF-FFT algorithm is shown in the figure below.

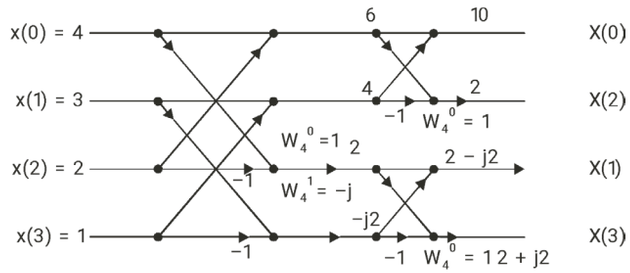


Fig. 3.22

$$Y(k) = \left(\underset{\uparrow}{10}, 2 - j2, 2, 2 + j2\right)$$

12. Compute the 8-point DFT of the sequence

$$x(n) = \begin{cases} 1 & 0 \le n \le 7 \\ 0 & \text{otherwise} \end{cases}$$

by using DIF-FFT algorithm. Show all intermediate results.

Solution:

Given that, N = 8, we know that $W_N^k = e^{-j\frac{2\pi}{N}k}$

Therefore,

$W_8^0 = 1, \quad W_8^1 = 0.707 - j0.707,$

$W_8^2 = -j, \quad W_8^3 = -0.707 - j0.707$

The butterfly diagram of an 8-point DIF-FFT algorithm is shown in the figure below.

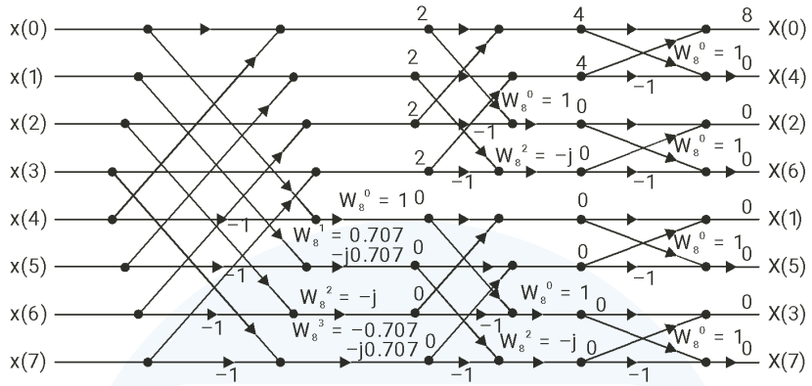


Fig. 3.23

$$X(k) = \left\{\underset{\uparrow}{8}, 0, 0, 0, 0, 0, 0, 0\right\}$$

13. Using the signal flow graph method DIF-FFT, determine the 8-point DFT of the sequence x(n) = {1,1,1,1,0,0,0,0}. Show the intermittent values on the graph.

Solution:

Given that N=8, we know that $W_N^k = e^{-j\frac{2\pi}{N}k}$

Therefore,

$W_8^0 = 1, \quad W_8^1 = 0.707 - j0.707,$

$W_8^2 = -j, \quad W_8^3 = -0.707 - j0.707$

The butterfly diagram of an 8-point DIF-FFT algorithm is shown in the figure below.

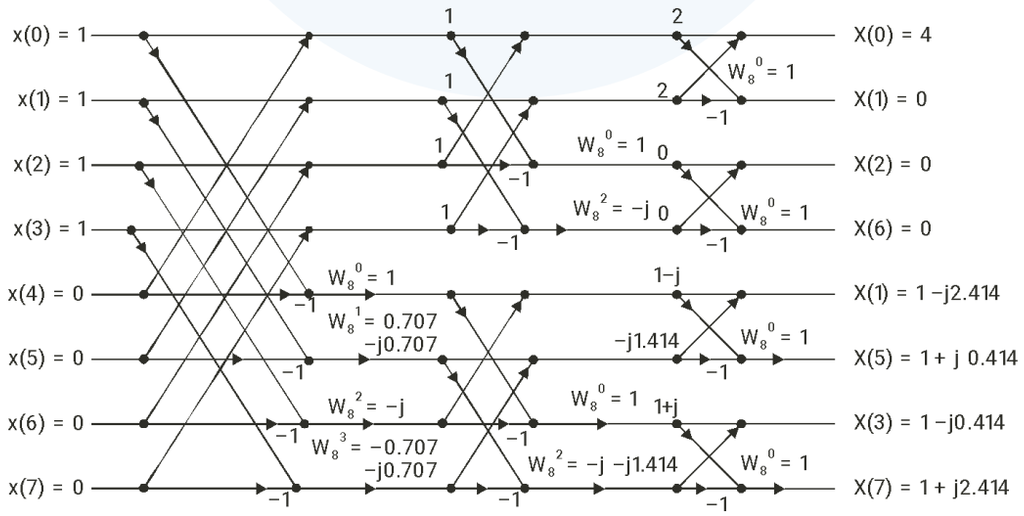


Fig. 3.24

$$X(k) = \left\{\underset{\uparrow}{4}, 1 - j2.414, 0, 1 - j0.414, 0, 1 + j0.114, 0, 1 + j2.114\right\}$$

14. Compute the 8-point DFT of the given sequence x(n) = {1,2,1,21,2,1,2} using DIF-FFT algorithm.

Solution:

Given that N=8. We know that $W_N^k = e^{-j\frac{2\pi}{N}k}$,

Therefore,

$W_8^0 = 1, \quad W_8^1 = 0.707 - j0.707,$

$W_8^2 = -j, \quad W_8^3 = -0.707 - j0.707$

The butterfly diagram of an 8-point DIF-FFT algorithm is shown in the figure below.

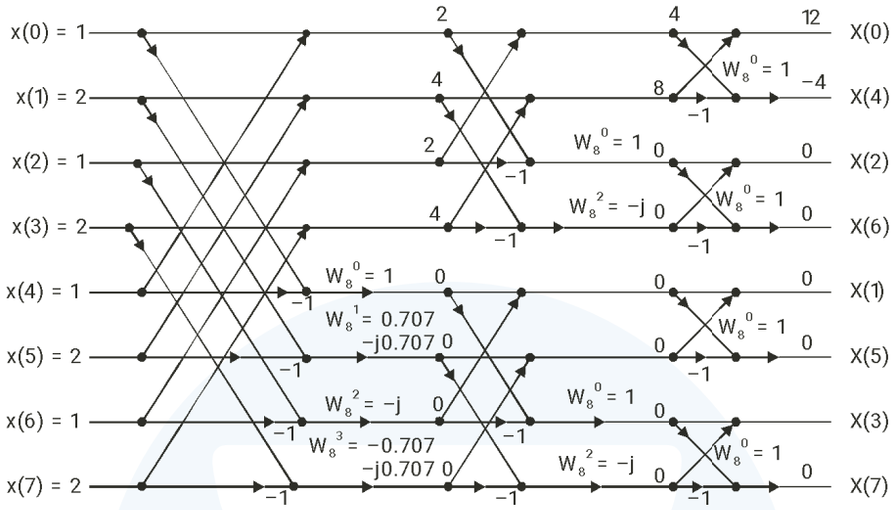


Fig. 3.25

$X(k) = \left\{\underset{\uparrow}{12}, 0, 0, 0 - 4, 0, 0, 0\right\}$

15. Compute the 8-point D.F.T of the following sequence, x(n) = {1, 2, 3, 2, 1, 2, 3, 2} using (a) DIT algorithm, (b) DIF algorithm,

Solution:

Given that N = 8. We know that $W_N^k = e^{-j\frac{2\pi}{N}k}$

Therefore,

$W_8^0 = 1, \quad W_8^1 = 0.707 - j0.707,$

$W_8^2 = -j, \quad W_8^3 = -0.707 - j0.707$

The butterfly diagram of an 8-point DIT-FFT & DIF-FFT algorithm is shown in the figure below.

$X(k) = \left\{\underset{\uparrow}{16}, 0, -4, 0, 0, 0, -4, 0\right\}$

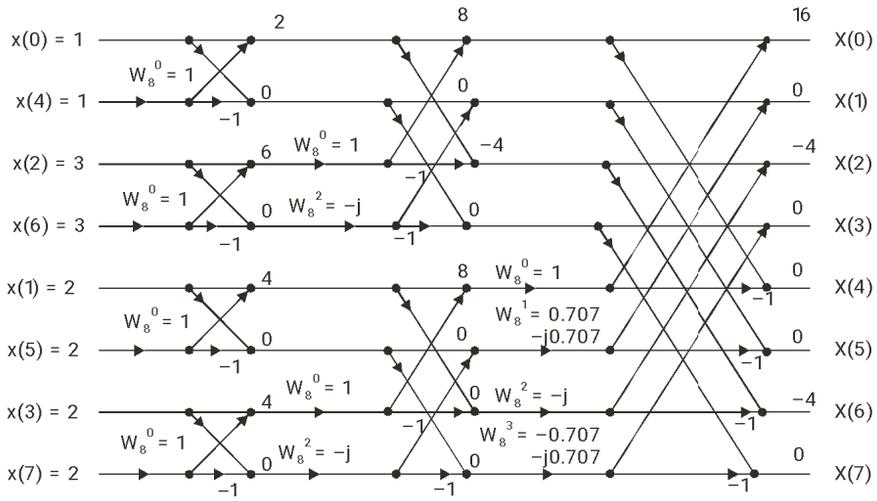


Fig. 3.26 Butterfly Diagram of an 8-point DIT-FFT Algorithm

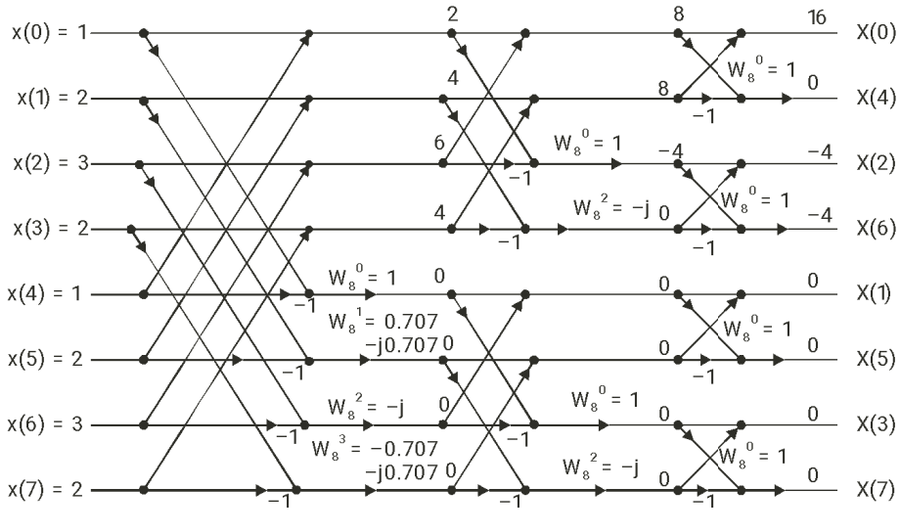


Fig. 3.27 Butterfly Diagram of an 8-point DIF-FFT Algorithm

Inverse DFT Using FFT Algorithms

Let X(k) be the N-point DFT of a length N sequence x(n). By definition of the inverse DFT:

$$x(n) = \frac{1}{N}\sum_{k=0}^{N-1} X(k) e^{j\frac{2\pi}{N}kn} = \frac{1}{N}\sum_{k=0}^{N-1} X(k)\left(e^{-j\frac{2\pi}{N}kn}\right)^*$$

$$= \frac{1}{N}\sum_{k=0}^{N-1} X(k)\left(W_N^{kn}\right)^*$$

$$x(n) = \frac{1}{N}\left[\sum_{k=0}^{N-1} X(k)\left(W_N^{kn}\right)^*\right]$$

In order to compute the inverse DFT from an FFT algorithm, following steps can be followed.

1. Take X(k) as the input sequence and z(n) as output sequence.
2. Compute FFT by replacing the twiddle factors W_N by.
3. Divide the output sequence by N.

If we consider the DIT-FFT signal flow graph, reverse its direction, change the sign powers of twiddle factors, and finally, divide the output by N. We obtain a DIF-FFT algorithm for computing the inverse FFT. This is shown in the figure below. Similarly, if we consider the DIF-FFT and repeat these changes, we get a DIT-FFT algorithm for computing the inverse FFT.

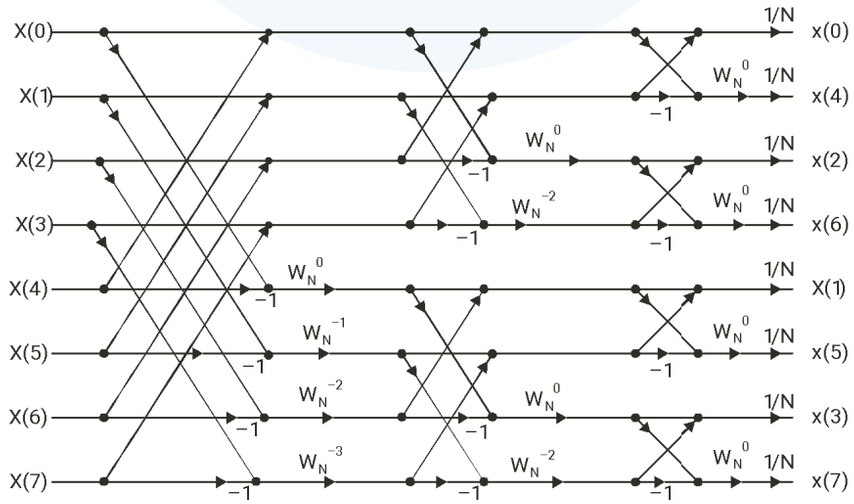


Fig. 3.28 Butterfly Diagram of an 8-point Inverse DIT-FFT Algorithm

Solved Examples

16. Compute the IDFT of the sequence X(k) = {7, −0.707 −j 0.707, −j, −0.707 − j0.707, 1, .0707 + j0.707, j, − 0.707 + j0.707} using DIF-FFT algorithm.

Solution:

Given that N = 8. We know that $W_N^{-k} = e^{\frac{j2\pi k}{N}}$ therefore,

$$W_8^0 = e^{-j\frac{(2\pi)}{8}0} = 1$$

$$W_8^{-1} = e^{j\frac{2\pi}{8}} = e^{j\frac{\pi}{4}} = \cos\frac{\pi}{4} + j\sin\frac{\pi}{4} = 0.707 + j0.707$$

$$W_8^2 = e^{j\frac{2\pi}{8}2} = e^{j\frac{\pi}{2}} = \cos\frac{\pi}{2} + j\sin\frac{\pi}{2} = j$$

$$W_8^{-3} = e^{-j\frac{2\pi}{8}3} = e^{j\frac{3\pi}{4}}$$

$$= \cos\frac{3\pi}{4} + j\sin\frac{3\pi}{4} = -0.707 + j0.707$$

The butterfly diagram of an 8-point inverse FFT algorithm in shown in the figure below.

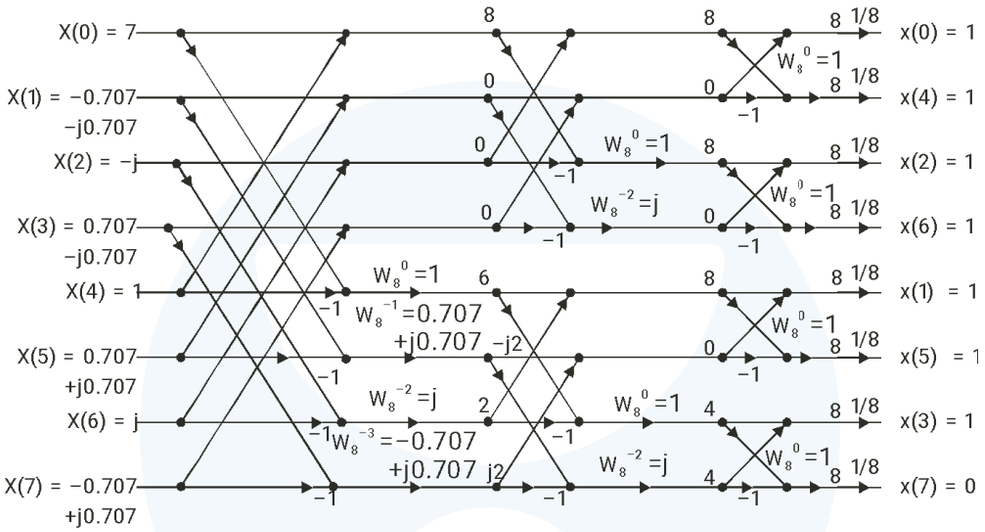


Fig. 3.29

$$x(n) = \left\{\underset{\uparrow}{1}, 1, 1, 1, 1, 1, 1, 0\right\}$$

17. Find the inverse FFT of the given X(k) = (1, 2, 3, 4) using DIF algorithm.

Solution:

Given that N = 4. We know that $W_N^{-k} = e^{j\frac{2\pi}{N}k}$

Therefore,

$$W_4^0 = 1,\ W_4^{-1} = e^{j\frac{2\pi}{4}} = e^{j\frac{\pi}{2}}$$

$$= \cos\left(\frac{\pi}{2}\right) + j\sin\left(\frac{\pi}{2}\right) = j$$

The butterfly diagram of a 4-point inverse FFT algorithm in shown in the figure below.

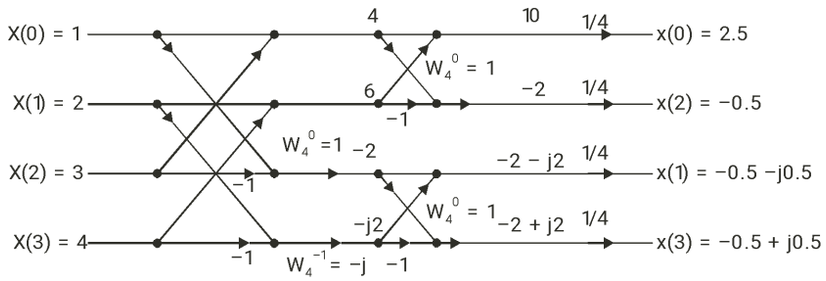


Fig. 3.30

$$x(n) = \left\{\underset{\uparrow}{2.5}, -0.5 - j0.5, -0.5, -0.5 + j0.5\right\}$$

18. Find the 4-point circular convolution of x(n) and h(n) given by x(n) = {1, 1, 1, 1} and h(n) = {1, 0, 1, 0} using radix-2 DIF-FFT algorithm.

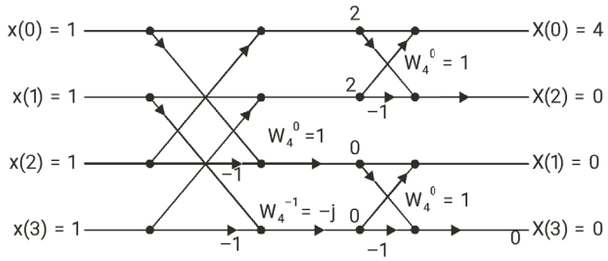
Solution:

Using Circular convolution property, we have

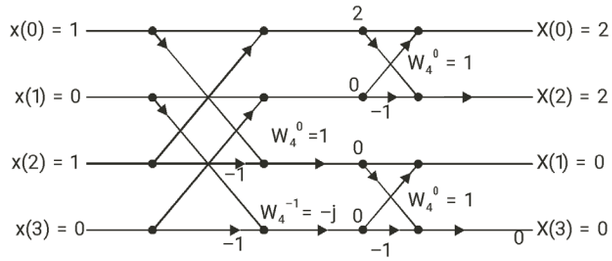
$y(n) = x(n) \textcircled{N} h(n)$.

$= x(n) \textcircled{N} h(n) \xrightarrow[N]{DFT} Y(k) = X(k)H(k)$

The butterfly diagram of a 4-point DIF-FFT algorithm for X(k) and H(k) is shown in Fig. 3.31(a) and 3.31(b) respectively. Fig. 3.31(c) shows the butterflies diagram to compute the inverse FFT y(n).



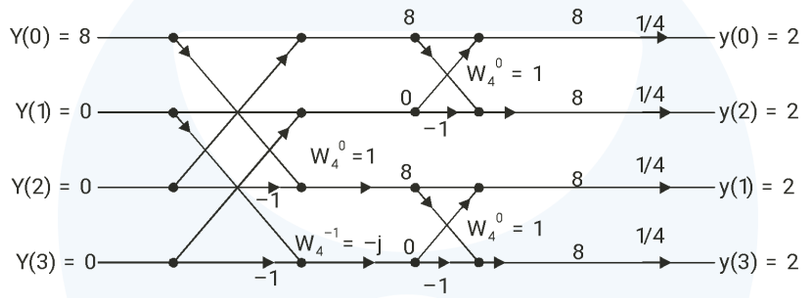
(a)



(b)

$X(k) = \left\{\underset{\uparrow}{4}, 0, 0, 0\right\}$

$H(k) = \left\{\underset{\uparrow}{2}, 0, 2, 0\right\}$



(c)

$y(n) = \left\{\underset{\uparrow}{2}, 2, 2, 2\right\}$

Fig. 3.31 Butterfly Diagram of a 4-point DIT-FFT Algorithm (a) for X(k), (b) for H(k), and (c) Butterfly Diagram of Inverse FFT for y(n)

Practice Exercise

Level-I

1. The algorithm used to compute any set of equally spaced samples of Fourier Transform on the unit circle is:
(A) DFT algorithm
(B) FFT algorithm
(C) Goertzel algorithm
(D) chirp transform algorithm

2. In radix 2-FFT algorithm, the value of N is:
(A) 2^m
(B) $2^{1/m}$
(C) 2m
(D) 2^{-m}

3. The number of computations required in DIT is _____ as that of DIF.
(A) different
(B) additional data required
(C) same
(D) double

4. What is incorrect about chirp transform algorithm?

(A) It is based on expressing DFT as a convolution.
(B) It is less flexible than FFT.
(C) It does not minimise computational complexity.
(D) It can be used to compute any set of equally spaced samples of the FT on the unit circle.

5. In an N-point FFT algorithm, _____ memory locations are required to store the coefficients.
(A) N – 3
(B) N^3
(C) $\frac{N}{3}$
(D) 3N

6. In DIT-FFT, _____ domain sequence is decimated.
(A) time
(B) frequency
(C) Both (A) and (B)
(D) None of the above

7. In DIF-FFT, _____ domain sequence is decimated.
(A) time
(B) frequency
(C) Both (A) and (B)
(D) None of the above

Level-II

1. The total number of complex additions required in radix-2 DIT-FFT algorithm is:
(A) $N \log_2 N$
(B) $\frac{N}{2}\log_2 N$
(C) $\frac{N}{\log_2 N}$
(D) None of the above

2. The total number of complex multiplications required in radix-2 DIT-FFT algorithm is:
(A) $N \log_2 N$
(B) $N\log_2 \frac{N}{2}$
(C) $\frac{N}{2}\log_2 N$
(D) $\frac{N}{2}\log_2 \frac{N}{2}$

3. A data sequence consisting of 32 samples is stored in memory in natural order. If the data sequence is subsequently scrambled using a bit-reversal algorithm, what will be the bit reversed indices for the data samples x(7) and x(13)? Assume a 32-point FFT processor is to be used.
(A) x(11) and x(14)
(B) x(14) and x(11)
(C) x(22) and x(28)
(D) x(28) and x(22)

4. The first four outputs of an 8-point radix-2 FFT are X(0) = 27, X(1) = –4 + j3, X(2) = 4 + j, X(3) = –j5. Which of the following statements is true?
(A) X(7) = j5
(B) X(7) = –4 –j
(C) dc value is 27
(D) X(7) = 5

5. In an 8-point DFT by radix-2 FFT, there are _____ stages of computations with _____ butterflies per stage.
(A) three, three
(B) four, four
(C) three, four
(D) four, three

6. The Goertzel algorithm is more efficient than FFT, when the DFT X(k) is computed for M values of k, where:
(A) $M \geq \log_2 N$
(B) $M \leq \log_2 N$
(C) M = N
(D) M > N

7. For an N-point FFT algorithm with $N = 2^m$, which one of the following statements is true?

(A) It is not possible to construct a signal flow graph with both input and output in normal.

(B) The number of butterflies in the mth stage is N/m.

(C) In-place computation requires storage of only 2N node data.

(D) Computation of a butterfly requires only one complex multiplication.

Answer Key

Level I

1. (A) **2.** (A) **3.** (C) **4.** (C) **5.** (D) **6.** (A) **7.** (B)

Level II

1. (A) **2.** (C) **3.** (B) **4.** (C) **5.** (C) **6.** (B) **7.** (D)